Audition & Speech Communication

Development of the Smart Speech™ serious game and the Smart Naso™ nasalance measurement apparatus for people with speech disorders (You et al., 2010 ~)

Contents

Part I

- Nature and Measurement of Sound
- Anatomy of the Ear & Hearing Process
- Fundamentals of Auditory Displays
- Principles for Auditory Display Design & Presentation

Part II

- Design of Auditory Displays
- Nature & Evaluation of Speech
- Factors Affecting Speech Intelligibility
- Synthesized Speech
- Quiz

Learning Goals (1/2)

GOALS

- Explain the primary **physical attributes of sound**, specifically **frequency and intensity**, and describe how they relate to the **human sensations of pitch and loudness**.
- Understand the concept of the **decibel (dB) as a logarithmic measure of sound pressure level (SPL)** and interpret how **changes in sound power** affect **decibel levels**.
- Differentiate between **pure tones and complex sounds**, and recognize how complex sounds can be represented using **sound spectra**.
- Identify the anatomical components of the **outer, middle, and inner ear**, and explain their distinct roles in transmitting sound vibrations to the brain.
- Compare and contrast the **place (resonance) theory and the temporal theory of hearing** to explain how the ear differentiates sound frequencies.

- Define the concept of **auditory masking** and its effects on **audibility**.
- Identify specific **environmental and task-related circumstances** where **auditory displays** offer significant advantages over visual displays.

Learning Goals (1/2)

GOALS

- Describe the four primary human functions involved in receiving auditory signals: **detection, absolute identification, relative discrimination, and localization**.
- Explain the concept of the **Just-Noticeable Difference (JND)** and its implications for designing auditory signals based on intensity and frequency discrimination.
- Explain how the human auditory system utilizes **phase and intensity differences to localize the direction of a sound source (stereophony)**.
- Apply **general design principles for auditory displays**, such as compatibility, approximation, and dissociability, to ensure effective information transfer.
- Recognize the **principles of presentation for auditory displays**, including avoiding extremes and establishing appropriate signal intensity relative to ambient noise levels.

Section 01

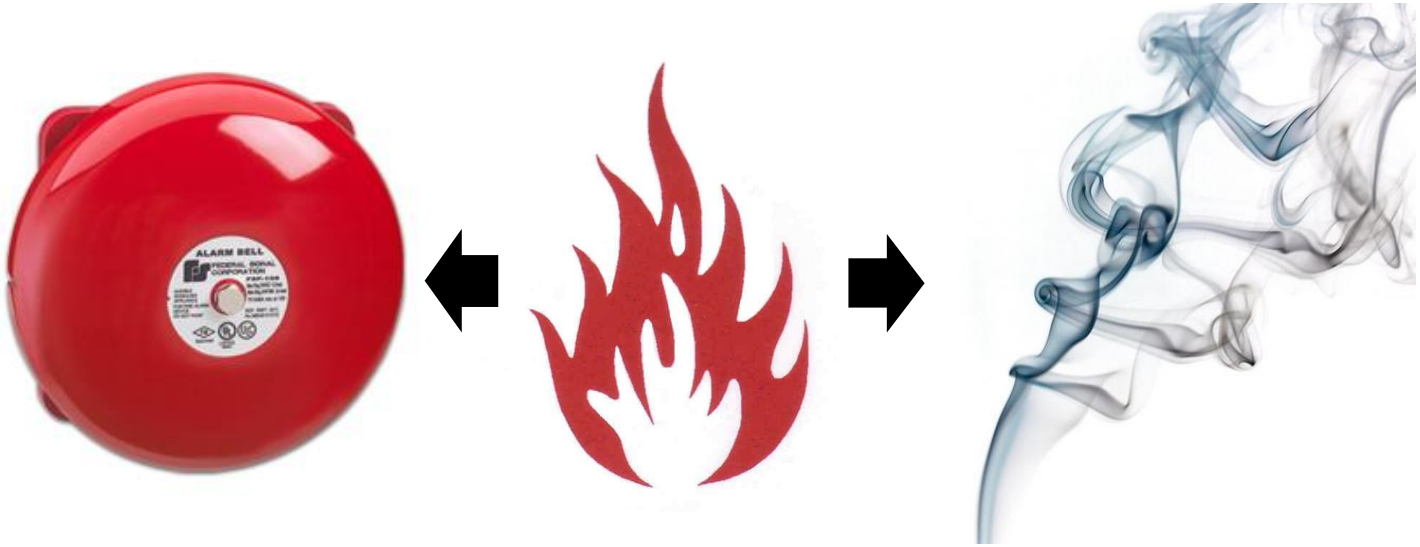
Nature & Measurement of Sound

Physical acoustics as a foundation for auditory display design

Effectiveness of Sound

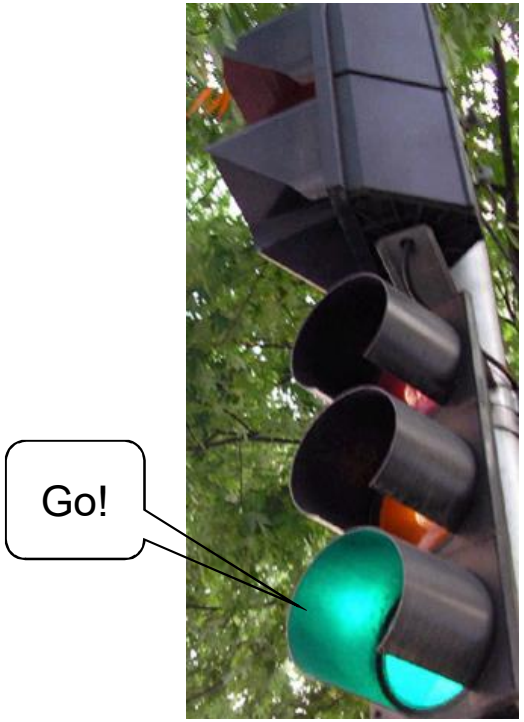
Sounds (alarms) can be MORE effective than direct stimuli. A fire alarm wakes sleeping subjects 100% of the time vs. 75% for heat/smoke alone (Kahn, 1983).

	Audio Alarm	Smoke, Smell, Heat
Recognition rate	100%	70%

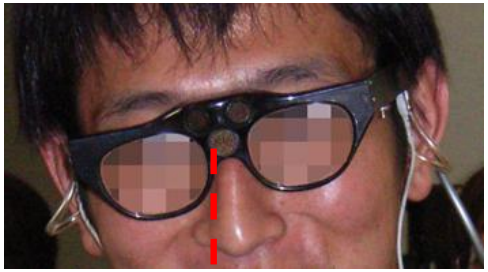


Utility of Sound

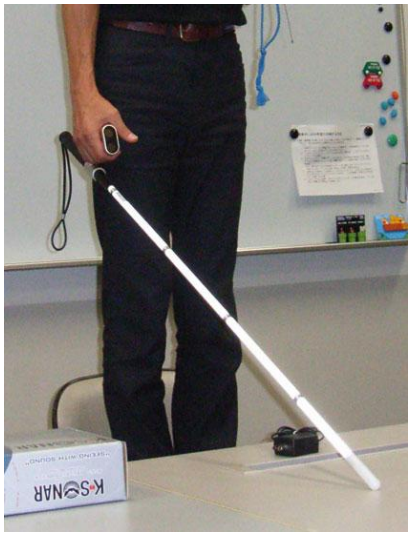
Clear-path indicator



Sonic guide

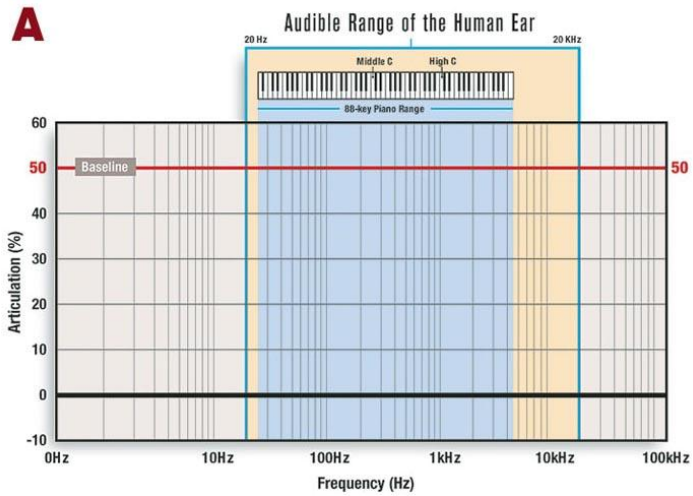


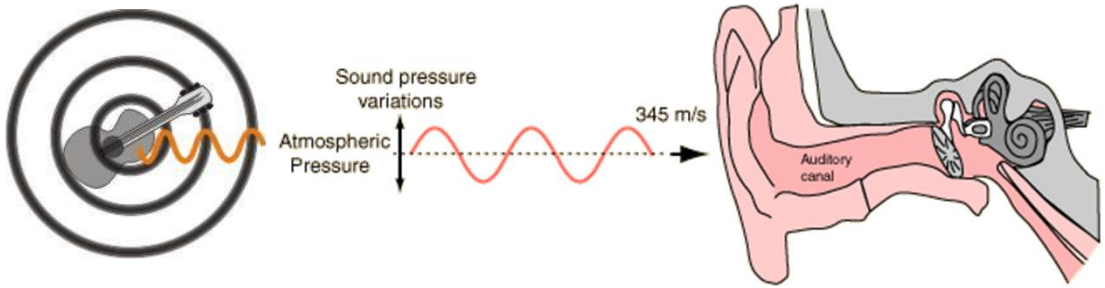
1) Distance
2) Direction
3) Surface characteristics

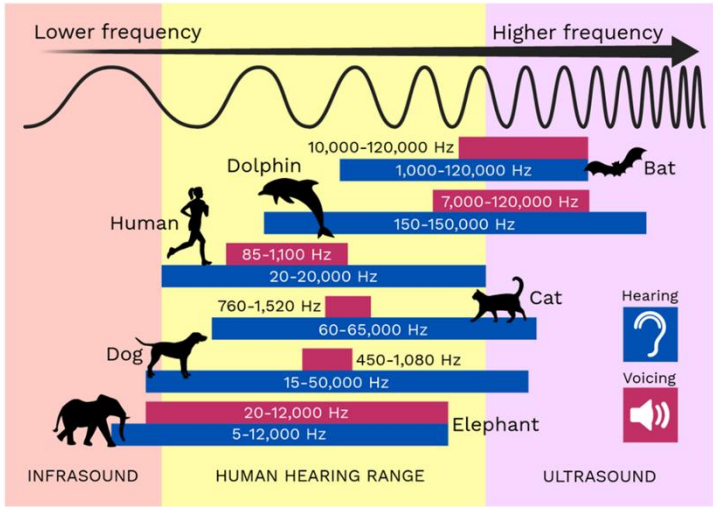


What is Sound?

- **Mechanical vibrations** transmitted through **air**
- Alternating compressions and rarefactions of air pressure
- Human hearing range: **20 ~ 20,000 Hz**
- Most real-world sounds are **complex (nonharmonic) mixtures**







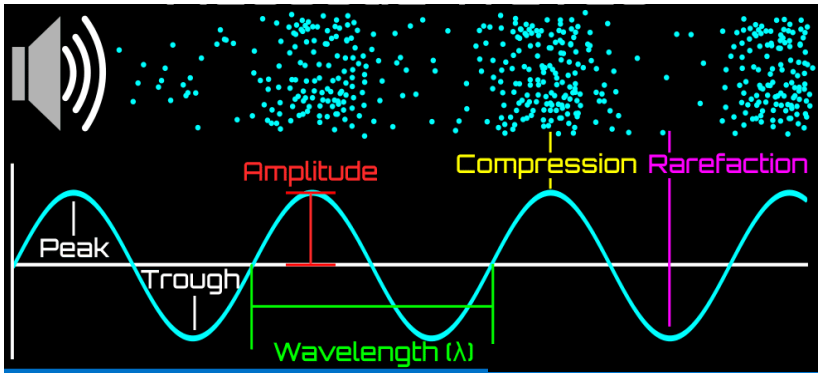
HF/E Significance of Sound

Physical Definition

Sound is a **mechanical wave** produced by vibration that propagates through a medium (usually air).

Vibration → Air pressure variation → Sine waveform

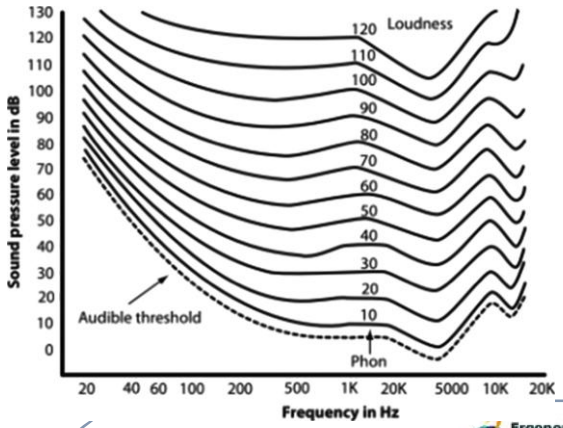
Two primary attributes
- **Frequency** (Hz): Cycles per second
- **Intensity** (W/m²): Power per unit area



HF/E Significance

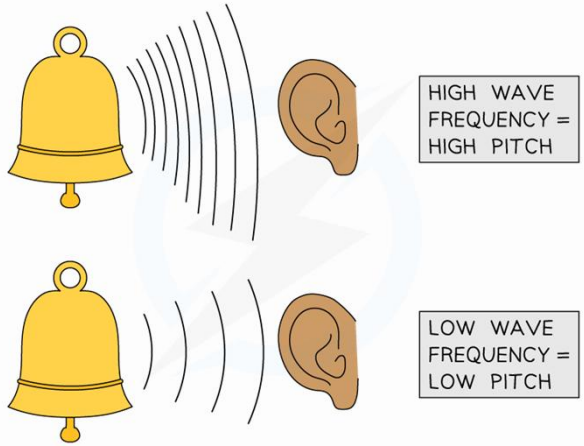
Sound serves as:
- **Warning & alarm** medium
- **Communication** channel (speech)
- Environmental **awareness cue**
- **Feedback signal** for system states

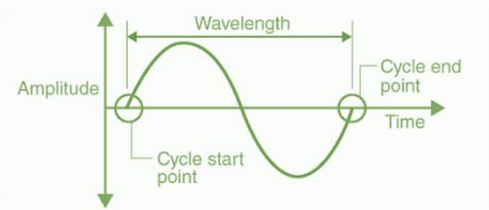
Two primary attributes
- **Pitch**: Perception of sound highness/lowness
- **Loudness**: Perception of sound intensity

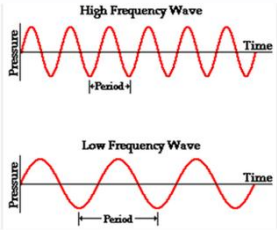


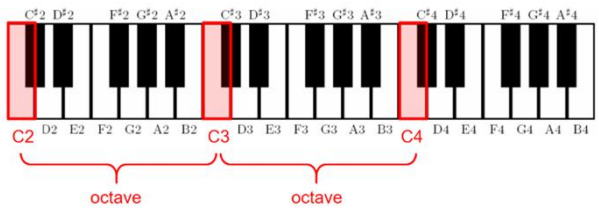
Frequency vs. Pitch

Aspect	Frequency	Pitch
Definition	The **physical measurement of the number of sound wave oscillations per second**.	The **subjective psychological perception or attribute of hearing** that allows sounds to be ordered on a scale from low to high.
Nature	**Objective and physical property** of sound waves.	**Subjective and perceptual attribute** existing within the auditory system.
Measurement Unit	Measured quantitatively in **Hertz (Hz)**.	Typically described qualitatively (**High/Low**) or via musical notation (e.g., **C4, A#**).
Determinant Factors	Determined solely by **frequency** of the sound.	Primary determinant is **frequency**, but is also influenced by **intensity** (loudness) and **duration** of the sound.
Measurability	Can be **measured precisely using electronic instruments** (e.g., an oscilloscope).	Difficult to measure directly; often assessed through perceptual matching tasks by **human subjects**.



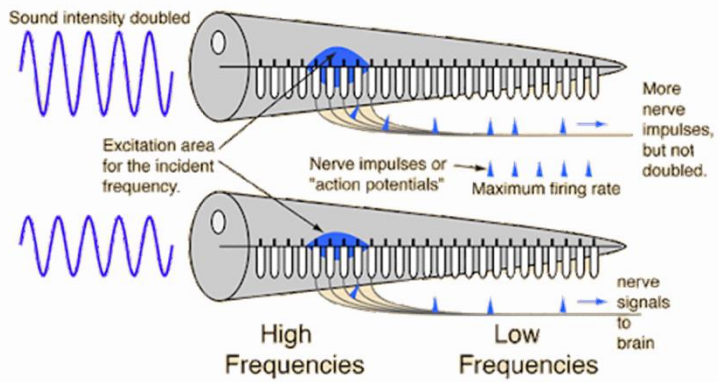


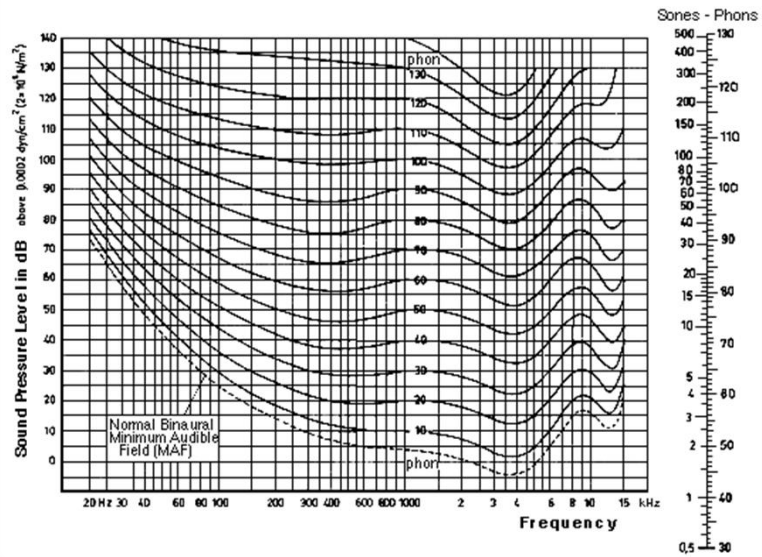




Sound Intensity vs. Loudness

Aspect	Sound Intensity	Loudness
Definition	The **physical measurement of the amount of sound energy passing through a specific area** per unit of time.	The **subjective psychological perception or sensation of the strength or magnitude of a sound** by a listener.
Nature	**Objective and physical property** of sound waves.	**Subjective and perceptual attribute** existing within the listener's auditory system.
Measurement Unit	Typically measured in **Watts per square meter (W/m²)**. Often expressed relatively in **Decibels (dB SPL/SIL)** using a logarithmic scale.	Typically described qualitatively (e.g., **Soft, Moderate, Very Loud**) or estimated using psychoacoustic units like **Sones** (perceived doubling) or **Phons** (at 1 kHz loudness level comparison).
Determinant Factors	Directly determined by **the amplitude (the maximum displacement) of the vibrating sound** source and the distance from the source.	Primarily determined by **sound intensity**, but is significantly influenced by the **frequency** of the sound, its **duration**, and the **presence of other sounds**.
Measurability	Can be measured precisely using calibrated electronic instruments, such as **sound level meters**.	Difficult to measure directly; often assessed through perceptual matching tasks by **human subjects**.





Sound Pressure Level (SPL) Formula

Sound Pressure Level (SPL) is a **logarithmic measure** of the effective pressure of a sound relative to a reference value (20 μPa).

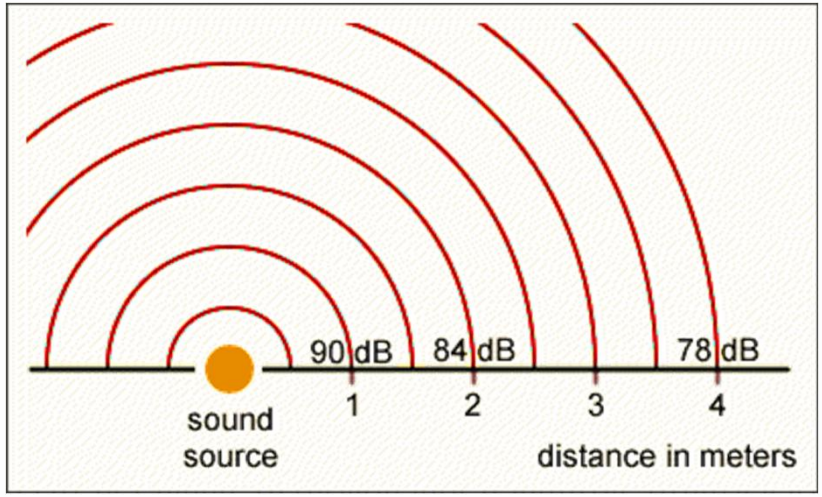
- Defined by power per unit area (W/m^2) ∝ pressure^2 (Pa = $1N/m^2$)
- It quantifies the physical intensity of a sound wave as perceived by human hearing, moving from absolute pressure variations to a usable, relative scale.
- When the **sound energy is doubled**, the total **SPL increases by 10 log2 (≈ 3) dB**.

SPL Formula

$$SPL(dB) = 10\log\frac{P_1^2}{P_0^2} = 20\log\frac{P_1}{P_0}$$

P_0 Reference pressure (20 μPa)

P_1 Pressure of sound being measured



Pure Tones vs. Complex Sound

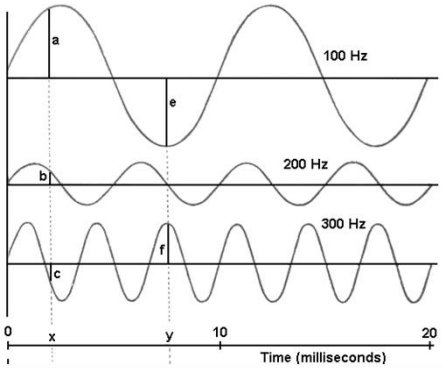
Pure Tone

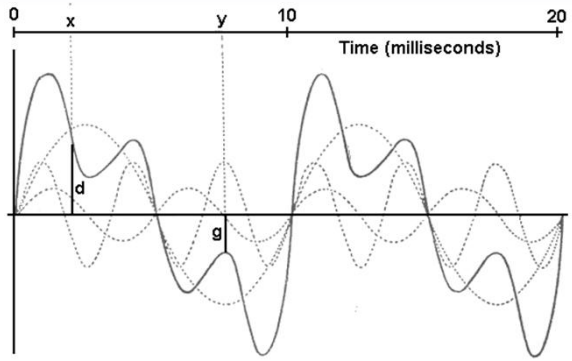
- Single-frequency sinusoidal wave
- Simple, clean sound quality (e.g., tuning fork)
- Spectrum shows a single vertical line
- Rare in natural environments; mostly lab-generated; useful in audiometric testing

VS

Complex Sound

- Composite of multiple frequency components
- Rich, textured sound (e.g., machinery, speech)
- Spectrum shows multiple lines or continuous bands
- Most real-world sounds are complex & nonharmonic

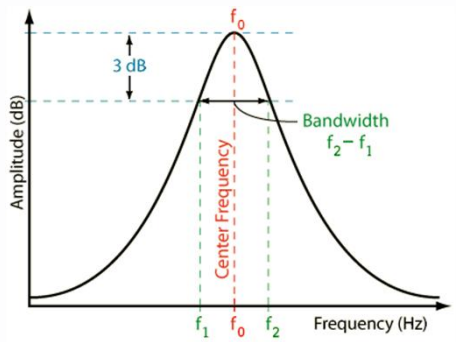




Bandwidth & Spectral Analysis

❑ Bandwidth

- Difference between the upper (f_2) and lower (f_1) frequency band limits
- Narrower bandwidth analysis yields greater spectral detail



❑ Center and band limit frequencies

- 1/N octave

$$f_1 = 2^{-\frac{1}{N}} f_C \qquad f_2 = 2^{\frac{1}{N}} f_C$$

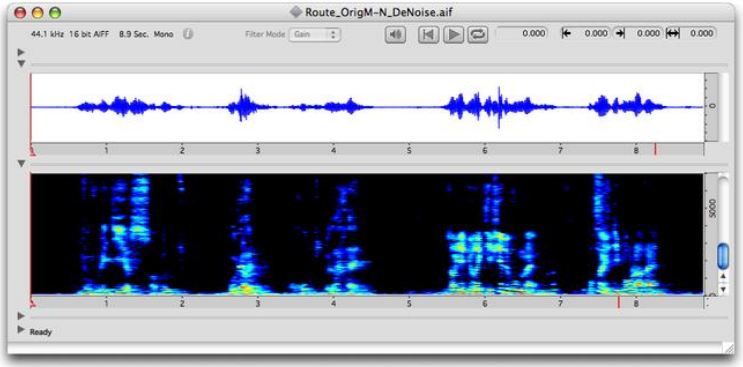
- 1/3-octave

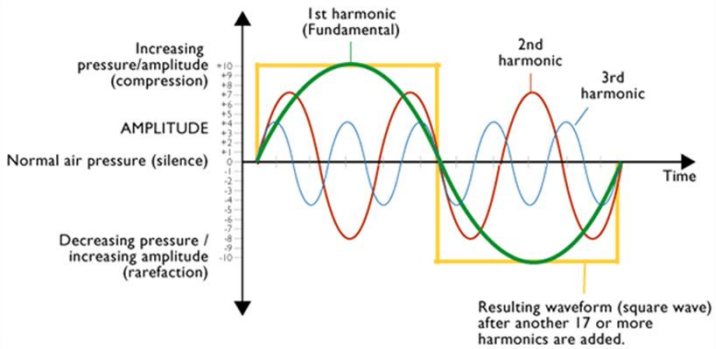
f_1	f_C	f_2
25	31.5	40
50	63	80
100	125	160
200	250	315
400	500	630

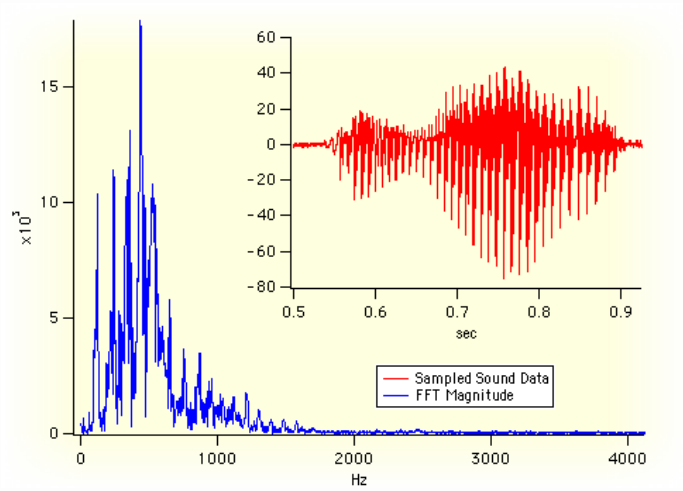
ANSI (1966) divides audible range into 10 bands.

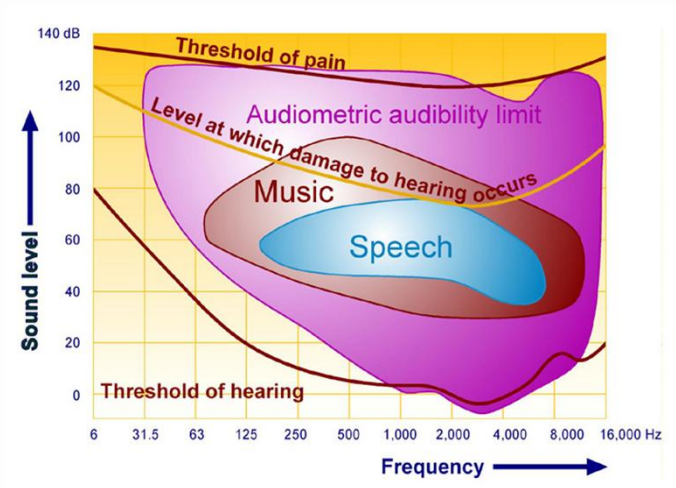
No	f_C	1/3-Octave Bands
1	31.5 Hz	25 Hz, 31 Hz, 40 Hz
2	63 Hz	50 Hz, 63 Hz, 80 Hz
3	125 Hz	100 Hz, 125 Hz, 160 Hz
4	250 Hz	200 Hz, 250 Hz, 315 Hz
5	500 Hz	400 Hz, 500 Hz, 630 Hz
6	1000 Hz	800 Hz, 1000 Hz, 1250 Hz
7	2000 Hz	1600 Hz, 2000 Hz, 2500 Hz
8	4000 Hz	3150 Hz, 4000 Hz, 5000 Hz
9	8000 Hz	6300 Hz, 8000 Hz, 10000 Hz
10	16000 Hz	12500 Hz, 16000 Hz, 20000 Hz

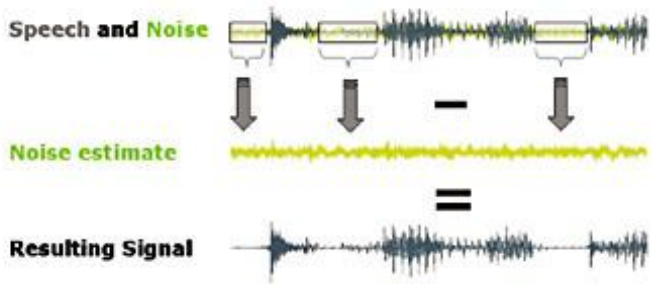
Complex Sound & Spectral Analysis







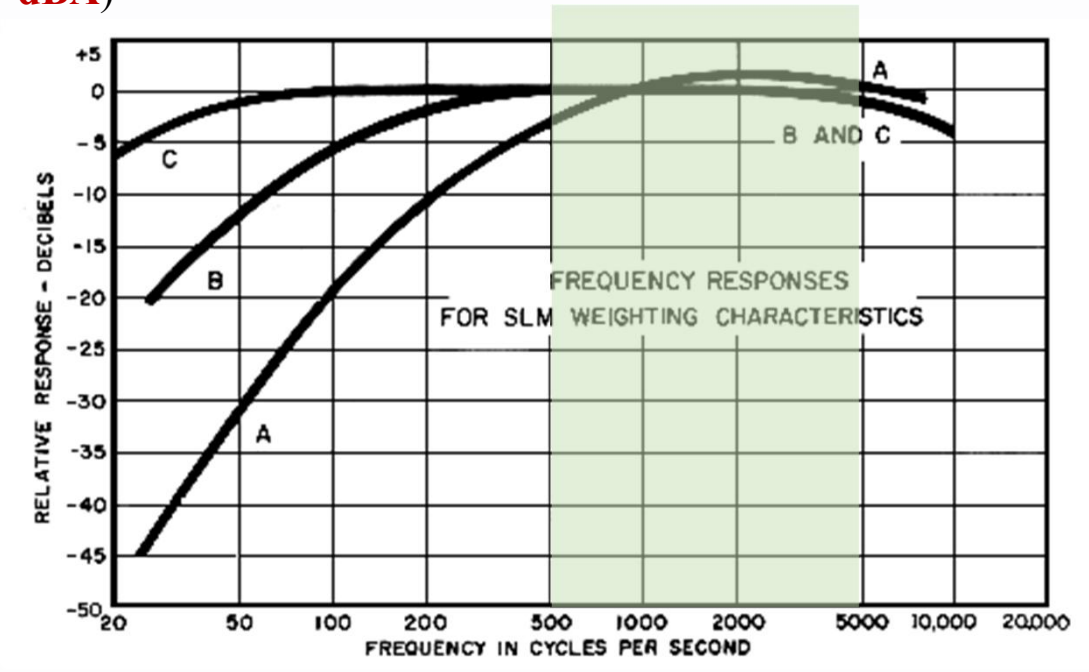




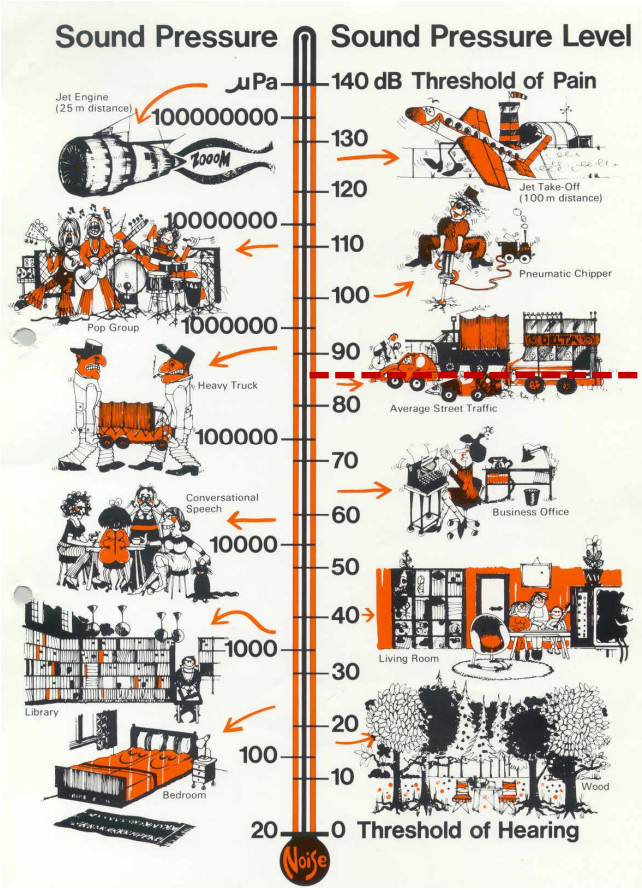
Sound Weighting Networks

❑ Frequency-Response Weighting Networks

- Established by ANSI (American National Standards Institute)
- Relative response curves of A, B, and C scales
- Many indexes of loudness, noisiness, and annoyance are **based on the A scale** (unit: **dBA; closest to the response characteristics of the human ear**; e.g., **occupational exposure limits for workplace noise = 85 dBA**)



Representative SPLs



Threshold of pain; rock concert, thunderclap;
Instant damage is possible

Busy street traffic, loud factory;
Prolonged exposure can cause hearing damage

85 dBA

Normal conversation

Quiet library or
residential area

Threshold of human hearing

Section 02

Anatomy of the Ear & Hearing Process

From sound wave to neural signal

Three Divisions of the Ear

Outer Ear

Structures

- Pinna (concha)
- Auditory canal (meatus)
- Tympanic membrane (eardrum)

Function

- Collects and channels sound energy
- Canal resonates at 2 ~ 5 kHz by up to 12 dB
- Eardrum vibrates with incident sound

Middle Ear

Structures

- Malleus
- Incus
- Stapes

Function

- Transmits vibration via ossicle chain
- Amplifies pressure ~22× at oval window
- Acoustic reflex (stapedius muscle) protects against intense sounds (protection latency 35 ~ 150 ms)

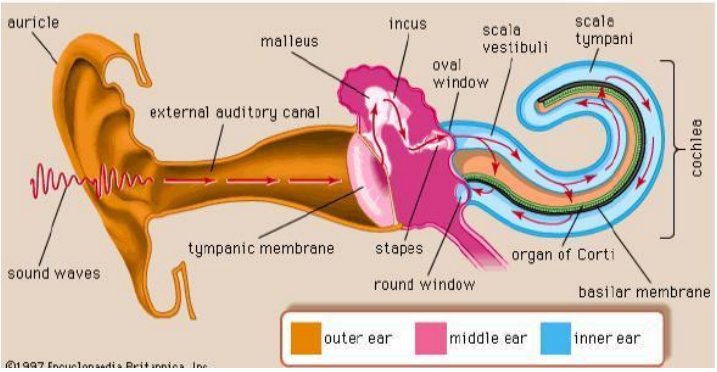
Inner Ear

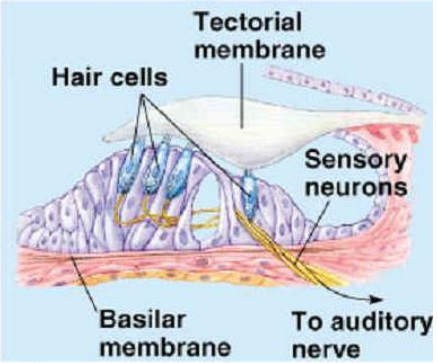
Structures

- Cochlea (spiral)
- Basilar membrane
- Organ of Corti
- Auditory nerve

Function

- Converts fluid vibration to neural impulses
- Hair cells highly sensitive to pressure changes
- Tonotopic organization

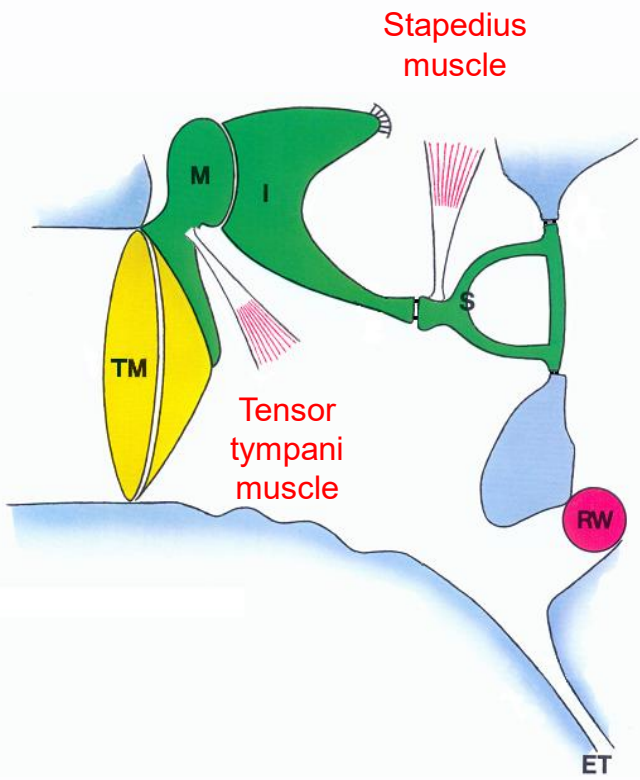




Acoustic (Aural) Reflex: Middle Ear Protection

How the ear protects itself from loud sounds

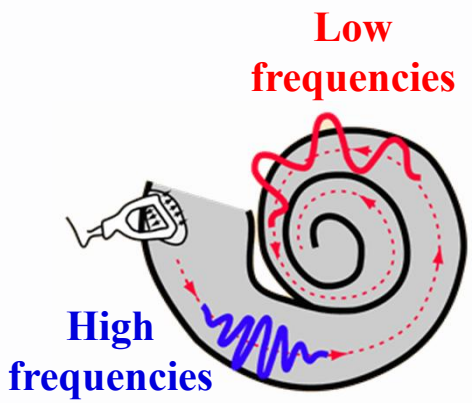
Trigger	Exposure to sounds ~80 dB above threshold; more responsive to broadband sounds and lower frequencies (Kryter, 1985)
Mechanism	Stapedius muscle tightens → reduces transmission from stapes to oval window
Attenuation	Up to 20 dB of protection from intense steady-state noise
Duration	Muscles remain flexed up to 15 min in steady noise; relax within ~0.5 s after an impulse
Latency Problem	35–150 ms delay before contraction → NO protection against the initial peak of impulse noise (e.g., gunfire)
Recovery	Relaxation after impulse: up to 2–3 s; subsequent impulses within 1 s still receive protection
Design Implication	Never rely on the acoustic reflex to protect against impulsive hazards; engineering controls and hearing protectors are required

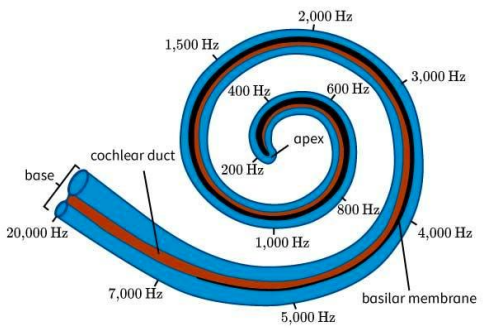


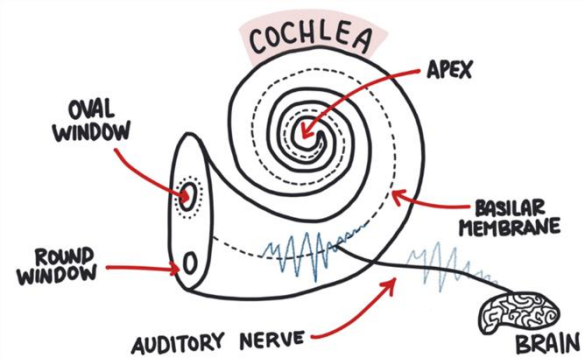
Theories of Hearing

Two complementary theories are currently accepted to understand the conversion of basilar membrane vibration into pitch perception.

Aspect	Place Theory (Location-Based)	Temporal Theory (Time-Based)
Mechanism	Sound waves cause peak vibration at specific **locations** on the basilar membrane.	The auditory nerve fires action potentials at **the time pattern of the incoming sound frequency**.
Frequency Matching	High-frequency sounds produce peak vibrations near the oval window, while low-frequency sounds vibrate the apex.	A 100 Hz sound causes neurons to fire 100 times per second, directly coding the pitch.
Best Explanation	Effective for explaining the perception of high-frequency sounds (1,000 ~ 5,000 Hz)	Effective for low-frequency sounds (< 1,000 Hz)
Limitation	Cannot fully explain the perception of very low-frequency sounds	Cannot explain high-frequency sounds, because neurons cannot fire fast enough (>1,000 times/sec)



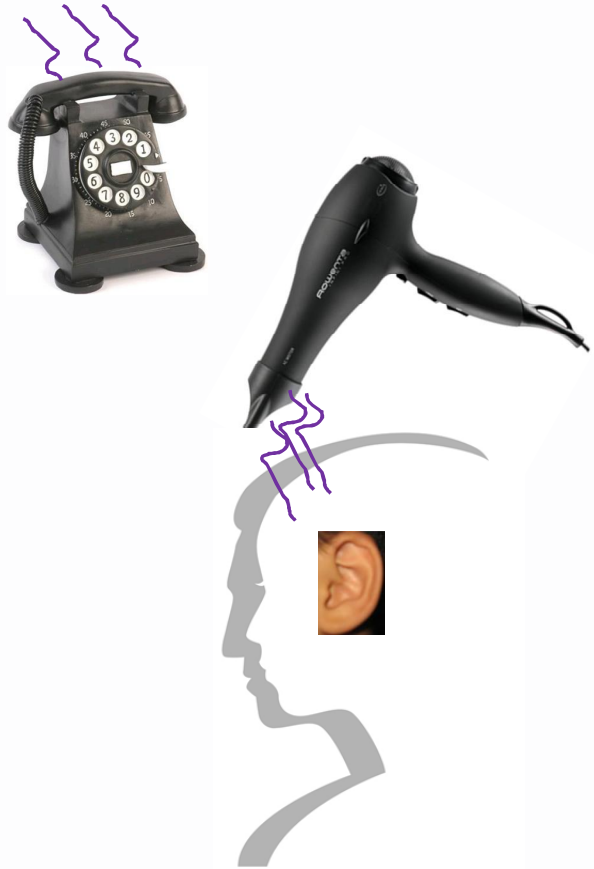




Auditory Masking

A psychoacoustic phenomenon in which **the threshold of audibility** of one sound (the masked sound) is **raised** by the presence of another sound (the masking sound).

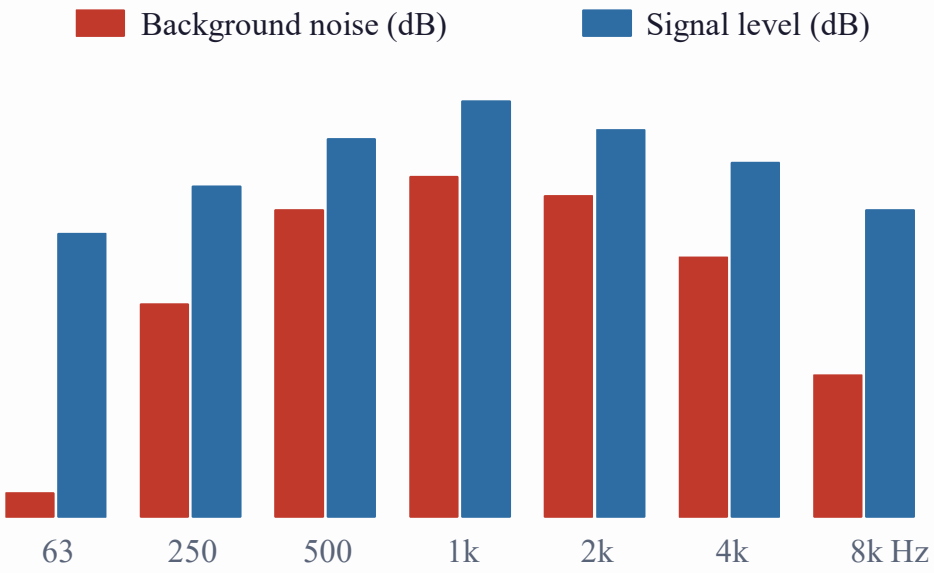
Near-frequency masking	Greatest masking occurs at and near the masking tone's frequency and its harmonics.
Intensity effect	Low-intensity maskers (20–40 dB) confine effect to nearby frequencies; high-intensity maskers (60–100 dB) spread to higher frequencies.
Narrowband vs. wideband noise	Narrowband masking resembles pure-tone masking; wideband masking depends on the 'critical band' around the masked tone.
Critical band	The width of the critical band increases with center frequency. Only noise within this band effectively masks the target tone.
Practical consequence	New York City (1974) had to abandon quieter car horn legislation — the quieter horns could NOT be heard over traffic noise (Garfield, 1983).



Masking: Design Implications

Understanding masking is essential for placing alarm signals in audible frequency windows.

Spectral Comparison: Background Noise vs. Signal Level



Design Rules

- ✓ Signal level must exceed masked threshold
- ✓ Min +15 dB above masked threshold (Patterson, 1982)
- ✓ Max +25 dB to avoid annoyance
- ✓ Place signal frequency in noise valleys
- ✓ Use complex signals (4+ frequency components) — harder to mask
- ✓ Measure background noise spectrum before specifying alarm

Section 03

Fundamentals of Auditory Displays

Use sound for detection, identification, discrimination, and localization

Situational Advantages of Auditory Displays

Use Auditory Presentation	Use Visual Presentation
Message is **simple** and **short**.	Message is **complex** and **long**.
Message will **not be referred to later**.	Message will be **referred to later**.
Message deals with **events in time**.	Message deals with **location in space**.
Message calls for **immediate action**.	Message does **not** call **for immediate action**.
Visual system is **overloaded**.	**Auditory** system is **overloaded**.
The location is **too bright or dark**-adaptation integrity is necessary.	The location is **too noisy**.
The receiver's job requires **moving about**.	The receiver's job allows them to remain in **one position**.

Four Auditory Functions

1 Detection

Determining whether a given signal is present.

Example: Is the fire alarm sounding?

2 Absolute Identification

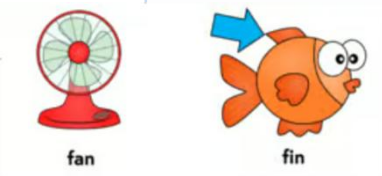
Identifying a particular signal when presented alone — without comparison.

Example: This is the 'system fault' alarm (not 'evacuation').

3 Relative Discrimination

Differentiating between two or more signals presented close in time.

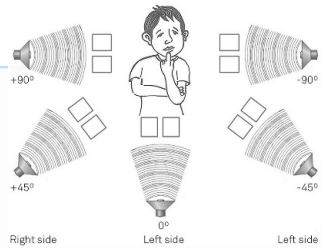
Example: Is this alarm the same as the last one?



4 Localization

Determining the direction from which a signal is coming.

Example: Is the siren behind or in front of the vehicle?



Detection of Auditory Signals

Key factors governing whether an auditory signal will be detected:

Ambient noise	Signal must exceed MASKED threshold, not just the absolute threshold. Always measure the noise environment first.
Signal intensity	In quiet conditions: 40–50 dB above absolute threshold is normally sufficient. In noise: follow the +15/+25 dB rule above masked threshold (Patterson, 1982).
Signal duration	The ear takes ~200–300 ms to 'build up' for pure tones. Signals < 200–500 ms sound less loud → should be longer, or intensity compensated.
Frequency	Detectability varies with frequency. Wideband sounds build up and decay faster than pure tones.
Startle effect	Very high-intensity signals can cause a startle reflex. This helps with simple reaction but DEGRADES choice reaction performance — moderate levels often better.
Filter-based enhancement	If noise and signal occupy different frequency bands, filtering out noise can raise effective S/N ratio without increasing signal level.

Methods for Signal Detectability in Noise

① **Reduce noise in the critical bandwidth** around the signal frequency.

② **Increase signal intensity** — but stay within **15 ~ 25 dB above masked threshold**.

③ Ensure **signal duration is at least 0.5 second** for pure tones.

④ **Shift signal frequency to a noise-valley band** where noise is relatively quiet.

⑤ **Phase-shift the signal**: present un-shifted to one ear, shifted to the other (CNS cancellation effect).

⑥ Present noise to both ears; **present signal to one ear only**.

⑦ **Use complex signals in the 1,000–4,000 Hz range** because complex signals are harder to mask.

Absolute Identification of Auditory Signals

People can identify far FEWER levels on a single dimension than they can discriminate in comparison. This is Miller's 'channel capacity' concept applied to auditory codes.

Auditory Dimension	Identifiable Levels	Engineering Significance
Intensity (pure tones)	4 ~ 5	Use only 4 distinct loudness levels in a coded system
Frequency	4 ~ 7	Limit tonal alarms to ≤7 distinct pitches for absolute ID
Duration	2 ~ 3	Short vs. long is the practical range for duration coding
Intensity + Frequency combined	9	Multidimensional coding yields more capacity

Multidimensional Coding Strategy

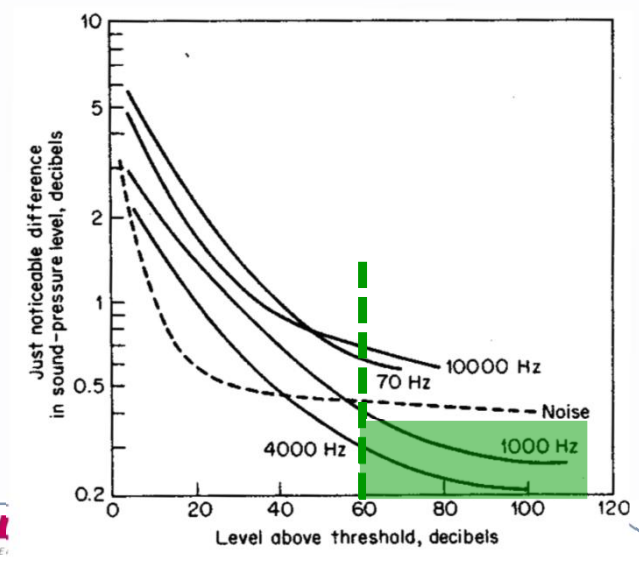
Use MORE dimensions with FEWER steps each (e.g., 8 dimensions × 2 levels) rather than fewer dimensions with more levels (e.g., 4 dimensions × 4 levels). This exploits the advantage of qualitative differences over quantitative differences (Pollack & Ficks, 1954).

Relative Discrimination: Just-Noticeable Difference (JND)

JND refers to the **smallest detectable change** in a stimulus dimension (intensity or frequency) that can be **noticed 50% of the time**. **Smaller JND** ⇒ **easier discrimination**.

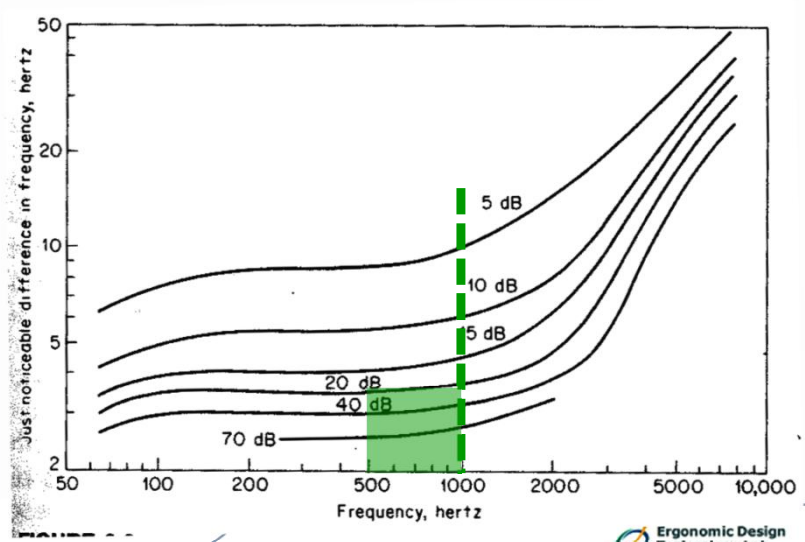
Intensity JND

- Smallest differences detectable at **≥ 60 dB** above threshold.
- Intermediate frequencies (**1,000 & 4,000 Hz**) yield smallest JNDs.
- Low-intensity signals have large JNDs → unreliable discrimination.



Frequency JND

- Smaller JNDs **below 1,000 Hz** (especially at high intensity).
- Signal intensity should be **≥ 30 dB** above threshold.
- Duration **> 0.1 s** required for reliable frequency discrimination.

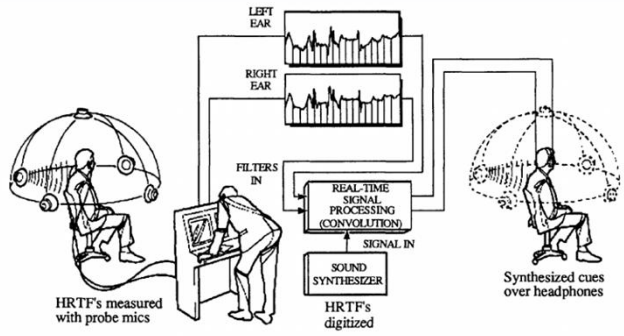


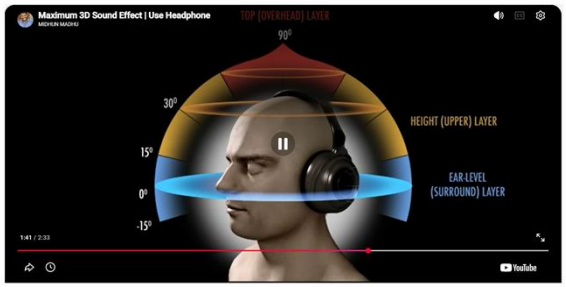
Sound Localization (Stereophony)

Interaural Time Difference (ITD)	Effective < 1,500 Hz	Sound reaches nearer ear ~0.8 ms earlier. Phase difference is an effective cue for lateral localization.
Interaural Level Difference (ILD)	Effective > 3,000 Hz	Head shadows the far ear; intensity differences become pronounced. Useful for left-right localization at high frequencies.
Spectral Cues (Pinna)	All	Pinna shape filters sounds differently based on elevation → used for up-down localization.
Head Movement	All	Turning the head resolves front-back ambiguity; critical for accurate 3D localization.

Head-Coupled Auditory Displays (Furness, 1986)

Manipulate interaural time difference (IRD) and interaural level difference (ILD) in real-time to stabilize sounds in 3D space as the pilot's head moves → enhances situation awareness, spatial cueing, and voice channel segregation.





Section 04

Principles for Auditory Display Design & Presentation

Principles considered for designing and presenting audio signals

Principles of Auditory Display Design

Compatibility

Exploit natural or learned relationships (e.g., high pitch = "up" or "high value"; faster rate = "urgent")

Approximation

Use two-stage signals for complex data: (1) Attention-demanding signal → (2) Designation signal with detail

Dissociability (Discriminability)

Ensure auditory signals are easily discernible from ongoing ambient noise, speech, or other alarms

Parsimony (Simplicity)

Provide only necessary information to the operator; avoid overwhelming cognitive channels

Invariance

The same signal should ALWAYS designate the same information across contexts

Standardization

Maintain consistent coding rules to facilitate learning and reduce errors

Principles for Audio Signal Presentation

- **Avoid extremes**: Prevent startle responses and annoyance that disrupt performance
- **Set intensity relative to ambient noise**: Calibrate intensity relative to ambient masked threshold (+15 to +25 dB).
- **Use interrupted signals**: Use variable (non-steady) signals to prevent perceptual adaptation and maintain attention
- **Don't overload**: Limit number of distinct alarms; test signals with representative users during installation